

British Society for Haematology updated guidelines for the diagnosis and management of tumour lysis syndrome in adults and children with haematological malignancies: A focus on patient safety

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KEYWORDS

clinical haematology, haematological malignancy, therapy

METHODOLOGY

This guideline update was compiled according to the British Society for Haematology (BSH) process at For Guidelines writers and Taskforce Members (b-s-h.org.uk). The Grading of Recommendations Assessment, Development and Evaluation (GRADE) nomenclature was used to evaluate levels of evidence and to assess the strength of recommendations. The GRADE criteria can be found at <http://www.grade-workinggroup.org>.

LITERATURE REVIEW DETAILS

A literature search was performed using the EMBASE and MEDLINE databases using the following search terms: 'tumour lysis syndrome' and 'tumor lysis syndrome' from 01/10/2013 to 01/06/2024. Papers published in non-English journals or non-human subjects were excluded from review.

REVIEW OF THE MANUSCRIPT

Review of the manuscript was performed by the BSH Guidelines Committee Haemato-oncology Task Force, the BSH Guidelines Committee and the Haemato-Oncology sounding board of BSH. It was also reviewed by the representatives from the National Health Service England (NHSE) Patient Safety team and Specialist Pharmacy Services (SPSs). These organisations do not necessarily approve or endorse the contents.

SCOPE

In response to a reported incident of delayed administration of rasburicase, the NHSE Patient Safety team undertook an analysis to answer the patient safety question, 'Are there wider issues relating to delayed or omitted administration of medication to treat/prevent tumour lysis syndrome?'. Between September 2016 and January 2020, the team identified 64 relevant reported incidents, of which 16

[Correction added on 2 October 2025, after first online publication: The copyright line was changed.]

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were reported as being associated with harm. The analysis identified several themes including a lack of staff knowledge (medical, nursing and pharmacy) around the management of tumour lysis syndrome (TLS), including the critical nature of medications to prevent/treat this complication. It also identified a lack of availability of critical medications to treat TLS at ward and pharmacy level.

Our writing group was formed in response to this analysis, to review the published literature and update the guidelines on the diagnosis and management of TLS in adults and children with a haematological malignancy with particular focus on patient safety. Where the guidance differs between adults and children, for example, medication doses or fluid infusion rates, this has been specifically stated in the text.

INTRODUCTION

TLS remains a life-threatening complication of haematological cancer and is caused when large-scale destruction of malignant cells releases intracellular contents and electrolytes into the bloodstream that overwhelm the normal physiological mechanisms of clearance. This can occur spontaneously in malignancies with a high proliferative rate (e.g. Burkitt lymphoma) but more commonly occurs following the administration of anti-cancer therapy.

Since the publication of the previous BSH guidelines for TLS in 2015,¹ there have been significant developments in the management of haematological malignancies with numerous novel targeted anticancer agents and immunotherapies. These represent important advances in therapy, but some are associated with a higher risk of TLS, including in diseases that previously would not have been considered high risk for TLS. We present an updated review of the management of TLS with particular focus on practical considerations to ensure patient safety and prompt drug delivery.

DEFINITION AND PATHOPHYSIOLOGY

The definition and pathophysiology of TLS remain unchanged from the 2015 guideline: Management of TLS in adults and children with haematological malignancies.¹

For completeness, we have reproduced the original definition of TLS² in Table 1 and TLS grading in Table S1.

RISK STRATIFICATION FOR TLS

Assessment of TLS risk is required for each patient prior to starting any new line of therapy. It is important that risk assessment considers patient and treatment factors in addition to disease risks. This combined approach will provide a personalised prophylactic strategy to mitigate the risk of TLS.

TABLE 1 Definition of tumour lysis syndrome (TLS).

Laboratory TLS The presence of two of more of the following abnormalities in a patient with cancer or undergoing treatment for cancer within 3 days prior and up to 7 days after initiation of treatment. • Uric acid $\geq 476 \mu\text{mol/L}$ or 25% increase from baseline • Potassium $\geq 6.0 \text{ mmol/L}$ or 25% increase from baseline • Phosphate $\geq 2.1 \text{ mmol/L}$ or 25% increase from baseline (children) $\geq 1.45 \text{ mmol/L}$ or 25% increase from baseline (adults) • Calcium $\leq 1.75 \text{ mmol/L}$ or 25% decrease from baseline
Clinical TLS A patient with laboratory TLS and at least one of: • Creatinine $\geq 1.5 \times \text{ULN}$ (age >12 years or age-adjusted) • Cardiac arrhythmia • Sudden death • Seizure

Source: Cairo and Bishop.²

TLS risk stratification requires consideration of the following factors:

1. Disease histology
2. Assessment of disease burden³
3. Patient-related factors, for example, renal dysfunction or renal involvement, frailty, ability to manage prophylactic strategies such as compliance with oral intake of fluids⁴
4. Assessment of pre-existing spontaneous laboratory or clinical TLS⁵
5. Treatment-specific therapeutic agents in certain diseases have a higher associated risk of TLS, for example, venetoclax in chronic lymphocytic leukaemia (CLL), chimeric antigen receptor T-cell therapy⁶⁻⁸

The disease-specific factors described in the international expert consensus panel guidelines⁹ are still relevant today and the TLS risk stratification flow charts from this paper are summarised with small modifications in Table 2. These highlight specific disease histologies and/or markers of higher disease burden, for example, lactate dehydrogenase (LDH)/disease stage/high white blood cell count (WBC)/cancer mass $>500 \text{ g}$ in adults or $>300 \text{ g/m}^2$ in children, that are associated with a higher risk of TLS.^{10,11}

Patient factors such as renal dysfunction and/or features of pre-existing TLS (e.g. hyperuricaemia/hyperphosphataemia/hyperkalaemia) will also influence the overall TLS risk assessment. For example, an adult patient with acute lymphoblastic leukaemia (ALL), WBC $<100 \times 10^9/\text{L}$ and normal LDH would be classified as intermediate risk for TLS on disease criteria alone. However, they would be 'upgraded' to high risk for TLS if they had co-existent renal dysfunction or other features of pre-existing TLS.

With the plethora of new therapeutic agents, further consideration needs to be given as to whether any of these 'upgrade' a patient's risk.⁶ Venetoclax in the treatment of CLL and acute myeloid leukaemia (AML) is a commonly used example, but it is not possible to provide a comprehensive list of all novel agents within this guideline. It is

TABLE 2 TLS risk assessment for haematological malignancies in adults and children based on disease and staging.

Disease		Modifying factors		TLS risk ^a
Myeloma		Alkylators/IMiD/PI/antibody		Low risk
		CAR T-cell and novel therapies including bispecific antibodies		Low–intermediate risk
Chronic leukaemias	Chronic myeloid leukaemia (chronic phase)			Low risk
	Chronic lymphocytic leukaemia	BTK inhibitors, alkylating therapies		Low risk
		Targeted and/or biological therapies e.g. venetoclax		Low–high risk
Acute leukaemias	Acute myeloid leukaemia	WBC <100 × 10 ⁹ /L		Intermediate risk
		WBC ≥100 × 10 ⁹ /L		High risk
	Acute lymphoblastic leukaemia	WBC <100 × 10 ⁹ /L		Intermediate risk
		WBC ≥100 × 10 ⁹ /L and/or bulky ^a disease		High risk
Lymphomas	Low-grade lymphomas e.g. Hodgkin/small lymphocytic/follicular/marginal zone/MALT/mantle cell (non-blastoid variant)/cutaneous T cell			Low risk
	High-grade lymphomas (B & T cell ^b) e.g. diffuse large B cell/mantle cell (blastoid variant or high Ki-67%)/transformed low grade lymphoma/anaplastic large cell lymphoma/ATLL	Early stage (I/II)	Normal LDH and non-bulky	Low risk
			LDH > ULN and/or bulky ^a	Intermediate risk
		Late stage (III/IV)	Normal LDH and non-bulky	Intermediate risk
			LDH ≥2× ULN and/or bulky ^a	High risk
	Burkitt lymphoma, lymphoblastic lymphoma	Early stage	LDH <2× ULN	Intermediate risk
			LDH ≥2× ULN and/or bulky disease ^a	High risk
		Advanced stage	High risk	

Note: The overall TLS risk can be further modified by patient factors, for example, presence of hyperuricaemia/hyperphosphataemia/hyperkalaemia or renal dysfunction/involvement.

Abbreviations: ATLL, adult T-cell leukaemia/lymphoma; BTK, Bruton tyrosine kinase; CAR T cell, chimeric antigen receptor T cell; IMiD, immunomodulatory drug; LDH, lactate dehydrogenase; MALT, mucosa-associated lymphoid tissue; PI, proteasome inhibitor; ULN, upper limit of normal; WBC, white blood cell count.

^aBulky disease is defined as a lymph node mass >10 cm in diameter on cross-sectional imaging. In paediatrics, there will be adjustments dependent on the child's size.

^bExcluding Burkitt lymphoma and lymphoblastic lymphoma.

vital for clinicians to keep up to date with published literature regarding the TLS risk in new therapies. Importantly, trial reported rates of TLS may differ from the incidence of TLS in the 'real-world' population.^{12–14}

Recommendations

- Every patient with a haematological malignancy commencing a new line of therapy should have a risk assessment for TLS. This should be personalised and include consideration of disease, patient and therapy factors (2B).
- Ongoing vigilance for tumour lysis development is required in all patients as it can subsequently develop even in patients initially deemed to be at low risk (2C).

PROPHYLAXIS OF TLS

Cases of TLS have been reported across a range of solid and haematological malignancies, even in those who are

receiving radiotherapy or non-systemic therapy such as intrathecal chemotherapy,¹⁵ and thus measures to reduce risk and incorporate monitoring strategies should always be considered. This includes those thought to be low risk.

Approaches to reduce the risk of TLS development include:

1. Hydration;
2. Uricosuric drugs—allopurinol, febuxostat, rasburicase;
3. Disease debulking, for example, corticosteroids pretreatment or dose escalation of therapy;
4. Patient education;
5. Avoidance of nephrotoxic drugs;
6. Monitoring.

Timing of the measures is also important, with the best risk reduction occurring if measures are put in place prior to definitive treatment of the malignancy, especially in those cancers which have a high risk of spontaneous TLS. Furthermore, decisions should be made as to whether a patient is suitable for outpatient strategies or requires hospital

admission for either treatment to be given intravenously or for more intensive monitoring strategies.

Hydration

Vigorous hydration and diuresis are a key part in preventing and managing TLS. Patients should be educated about the importance of drinking 2–3 L/day of fluids if possible depending on renal function, equivalent to 2.5–3 L/m²/day in children, starting the day prior to the treatment. If necessary, intravenous hydration should be used to meet these targets, particularly in those at high risk for developing TLS.

Loop diuretics could be considered in those liable to fluid overload, as this promotes potassium excretion through urine and has fewer interactions (e.g. with allopurinol) than other diuretics.

Potassium and phosphate supplementation should be avoided.

Caution is required in patients with high count leukaemia and low haemoglobin, as hyperhydration may cause further haemodilution and exacerbate anaemia.

There is no evidence that alkalinisation of urine reduces the risk of TLS as, while it increases the solubility of uric acid, it does not increase the solubility of xanthine and hypoxanthine. Alkalinization decreases the solubility of Ca²⁺ and PO₄⁻, increasing the development of calcium/phosphate crystals and renal dysfunction (Figure 1).

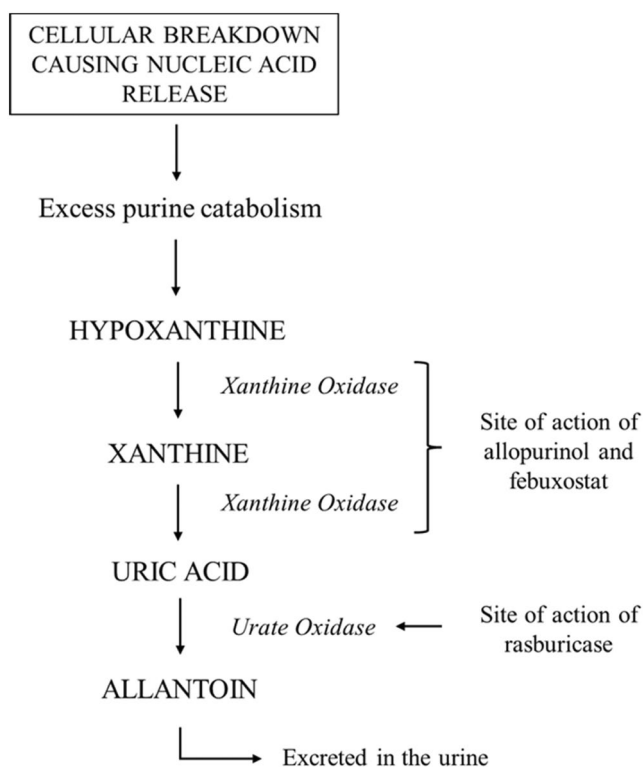


FIGURE 1 Mechanisms of action of xanthine oxidase inhibitors (allopurinol and febuxostat) and exogenous urate oxidases (rasburicase).

Uricosuric drugs

The choice of uricosuric drug depends not only on the TLS risk of the patient, but also on the urgency with which definitive therapy needs to commence, potential toxicities of the agents and pre-existing uric acid and phosphate levels. One should also consider the dose of the uricosuric agent to give (Table 3).

Allopurinol can be considered in those with low or intermediate risk of TLS. The usual dose administered is 300 mg daily orally although higher doses can be given and there is an intravenous formulation (although not often routinely stocked). This should be started at least 12 h prior to definitive therapy. For children, allopurinol should be given at a dose of 100 mg/m² 8-hourly. Caution should be employed in those with pre-existing renal impairment or in those who are taking diuretics or other drugs that may potentially interact with allopurinol. Prescribers should also be aware of the high incidence of skin reactions with allopurinol.¹⁶

An alternative xanthine oxidase inhibitor with fewer interactions is febuxostat 120 mg daily for adults. The safety and efficacy of febuxostat in children under the age of 18 years has not been established and not currently recommended in this guideline.

A phase III double-blind study randomising patients between febuxostat and allopurinol showed a more effective reduction in uric acid levels in those taking febuxostat although there was no difference in TLS rates between the two arms.¹⁷ A lower dose of 60 mg has been trialled successfully in a randomised non-inferiority study with allopurinol in the control arm for 100 patients with an intermediate or high risk of TLS.¹⁸ Allopurinol and febuxostat have demonstrated similar efficacy in retrospective studies of 45 paediatric patients¹⁹ and 78 adult patients.²⁰ There is real-world evidence for the use of even lower doses, for example, 10 mg, but these studies involve a very small number of patients and, at this point, this approach cannot be recommended outside a clinical trial.²¹ It should be noted that there have been safety reports regarding hypersensitivity reactions with febuxostat, and the Medicines and Healthcare products Regulatory Agency (MHRA) cautions its use in those with pre-existing major cardiovascular disease.²²

There have been cases of xanthine crystal formation leading to xanthine nephrolithiasis and worsening renal function in patients receiving allopurinol and febuxostat.^{23,24} This is thought to occur as xanthine is less soluble than uric acid. Xanthine levels are not routinely measured in hospital laboratories but, if this complication is suspected, management would include hydration, discontinuation of the xanthine oxidase inhibitor and switching to rasburicase.

We recommend the use of rasburicase for patients at high risk of TLS, or in those who are at intermediate risk and cannot take xanthine oxidase inhibitors, or in whom immediate therapy is required and pre-existing urate levels are high.

Patients who may receive rasburicase for either TLS prophylaxis or treatment should be tested for glucose

TABLE 3 Uricosuric drugs used for the prevention of tumour lysis syndrome (TLS) in haematological malignancies.

Drug	Indication	Adult dose	Paediatric dose	Additional information
Allopurinol	Low–intermediate risk TLS	300 mg OD PO	100 mg/m ² 8-hourly	Skin reaction. Dose reduction in renal impairment. Interaction with diuretics
Febuxostat	Low–intermediate risk TLS and unable to take allopurinol	120 mg OD PO	Not recommended in children	Hypersensitivity reactions. Caution if co-existing cardiovascular disease
Rasburicase	High-risk TLS or intermediate risk TLS and unable to take allopurinol or febuxostat	0.2 mg/kg OD licensed dose A single fixed dose (3 mg) may be sufficient for prophylaxis	0.2 mg/kg OD	Affects uric acid levels ex vivo leading to falsely low urate levels unless samples kept on ice. Avoid in G6PD deficiency due to the risk of haemolytic anaemia and methaemoglobinemia

Abbreviations: kg, kilogram; mg, milligram; OD, once a day; PO, oral administration; TLS, tumour lysis syndrome.

6-phosphate dehydrogenase (G6PD) deficiency, as rasburicase administration in at-risk patients can result in haemolytic anaemia or methaemoglobinemia. This should ideally happen at the time of diagnosis, in all patients, but particularly in those whose ethnic origin suggests a significant probability of deficiency, that is, in people of African, Mediterranean or Asian descent.²⁵ However, this may be challenging to achieve in practice, with one study demonstrating only 18% of patients had G6PD historical screening prior to the administration of rasburicase.²⁶

The licensed dose of rasburicase in adults and children is 0.2 mg/kg/day for 5 days as an intravenous infusion over 30 min; although there have been many low-quality studies exploring either single dose²⁷ or single fixed dose strategies. As per previous guidelines, in adults we recommend that a single fixed 3 mg dose of rasburicase may be sufficient as prophylaxis, with appropriate monitoring and a further 3 mg dose given if necessary.^{28,29} Lower doses have been trialled in small case series.³⁰ If a patient develops clinical TLS on subsequent monitoring, then they should be treated as having established TLS, which includes increasing the dose of rasburicase to 0.2 mg/kg/day for up to 7 days in adults or switching to rasburicase if patients were previously on either allopurinol/febuxostat. Escalating rasburicase to treatment dose should also be used for patients who develop biochemical signs of TLS on subsequent monitoring.

In paediatrics, we continue to recommend using the licensed dose of 0.2 mg/kg for up to 5 days for prophylaxis in patients who are high risk for TLS.³¹

Rasburicase can affect uric acid levels ex vivo, leading to falsely low urate levels and ideally samples post-rasburicase should be sent to the laboratory on ice to prevent this from happening if uric acid levels are required for clinical decision-making.

It should be noted that while there is clear evidence that rasburicase reduces the level of uric acid more quickly than allopurinol, this has not translated into a lower rate of renal impairment, albeit based on limited low quality evidence.^{32–35}

If rasburicase is given initially as a single dose, continuation of prophylaxis with allopurinol can be considered and can be recommenced 24 h after last rasburicase dose.³⁶

Disease debulking strategies

There is evidence in those high risk for TLS either due to inherent disease properties such as ALL or in those who are having therapy that increases their risk for TLS for example, venetoclax for CLL, that by treating beforehand either with corticosteroids or alternative therapy such as obinutuzumab or ibrutinib, the rate of TLS reduces substantially.³⁷ Similarly, stepwise dose escalation of those treatments that markedly increase risk has also been employed to reduce severity of TLS.

Retrospective evidence for this strategy was demonstrated in paediatric patients with ALL in two tertiary care hospitals in South Korea. By comparing the incidence of TLS in those who had prephase steroid treatment for at least 7 days compared to those that did not, they found an 88% reduction in the risk of TLS in those who had steroids.³⁸ In contrast, debulking strategies for AML have not been shown to be beneficial, but this may perhaps be due to the differing baseline risk of TLS between the two types of leukaemia.³⁹

While there may be a theoretical rationale for leukapheresis reducing the risk of TLS in those with newly diagnosed acute leukaemia with leucocytosis, this has not been borne out by retrospective analyses.^{39,40} We do not recommend this strategy for reducing TLS risk due to lack of evidence.

Patient education

Patients should be counselled on the risk of TLS and the relevant signs of symptoms prior to starting therapy. The need for good oral fluid intake and advice to seek medical attention should they develop reduced urine output should be emphasised. Education should also include the importance of adherence to the prophylactic regimen and outpatient monitoring schedule. This may include a need to stay local to the treating centre for a specified period of time. In some cases, this may require review at a specialty centre.

Avoidance of nephrotoxic drugs

A thorough review of the patient's existing medication should be undertaken to consider those that may potentiate

electrolyte abnormalities, for example, amiloride and other potassium-sparing diuretics, those that may affect renal vasculature such as non-steroidal anti-inflammatory drugs, angiotensin-converting enzyme inhibitors/angiotensin receptor blockers, sodium-glucose cotransporter-2 inhibitors or other potentially nephrotoxic drugs, for example, metformin.⁴¹

Monitoring

Monitoring forms a key part of TLS prophylaxis and should be tailored to predicted risk. Blood tests, including renal function, electrolyte levels (i.e. potassium, calcium, phosphate) and uric acid or urate levels, should be checked pre-treatment and at least twice daily in patients at high risk of developing TLS, increasing to up to four times a day in those who are at very high risk, for example, those patients who have high-risk disease and multiple modifying factors that also increase risk for TLS.

Patients with high risk for TLS who are admitted should have strict fluid balance monitoring, ensuring a good urine output.

In addition to the recommendations above covering general principles, some therapeutic agents with a high risk of TLS have their own suggested strategies to assess and mitigate the risk, for example, venetoclax in CLL.^{42,43}

Recommendations

- Patients should be encouraged to drink 2–3 L fluid (children: 2–3 L/m²/day) the day before, the day of and the day after initiation of a new line of therapy. If patients are at high risk for TLS or are unable to consume the recommended amount, then they should be considered for intravenous fluid (2C).
- All patients should be educated about the importance of adequate hydration, avoidance of nephrotoxic drugs and the signs of symptoms of TLS (2C).
- Patients with low risk of TLS: consider allopurinol for the first 7 days of the first cycle of treatment (2C).
- Patients with high risk of TLS: consider intravenous hydration. Consider prophylactic rasburicase, 3 mg may be sufficient but may need to be repeated depending on blood tests. Bloods should be monitored at least twice daily but may be required up to 3–4× per day depending on the clinical situation (2C).
- Patients with intermediate risk of TLS: Oral hydration and allopurinol may be sufficient for these patients but must be considered on a case-by-case basis, as some may warrant a strategy as per high-risk patients (2C).
- Prophylactic uricosuric drugs should be prescribed and administered prior to the treatment for the haematological malignancy (2C).
- Debulking strategies may be helpful in reducing the risk of TLS, such as prephase corticosteroids in patients with DLBCL (2C).

- Ensure clear communication to out of hours/on call teams regarding patients who are at risk of TLS or are being treated for TLS. This is particularly important when patients are located in clinical areas less familiar with the management of haematological malignancies and TLS (2C).

TREATMENT OF ESTABLISHED TLS

A multidisciplinary approach is required to manage patients with established TLS and should involve adult or paediatric haematologists, nephrologists and intensive care physicians to ensure optimal patient outcome. It is recommended that patients are managed in an inpatient setting where the wider team (including nursing staff, health care assistants and pharmacists) have experience in managing haematology patients and the complications of disease and chemotherapy.

Hydration and fluid balance

Vigorous hydration with intravenous fluid is an essential component of TLS treatment. For adults, a rate of 3 L/m² should be initiated and the ongoing rate adjusted to maintain a urine output of 100 mL/m²/h. In children, a rate of up to 4 L/m² can be given, depending on renal function, clinical and fluid status.

Electrolyte disturbances

In cases of TLS, the management of electrolyte disturbances should follow local guidelines. In cases of hyperkalaemia, this may include administration of nebulised salbutamol, intravenous insulin and glucose solutions or oral potassium binders such as patiomer, sodium zirconium cyclosilicate and calcium resonium. Potassium must not be added to any intravenous fluids administered. Caution should be exercised in the administration of balanced crystalloid solutions, for example, Hartmann's, Plasma-Lyte, as they also contain potassium.

Hypocalcaemia should be treated if a patient is symptomatic, for example, cardiac arrhythmias, seizures or tetany. Hyperphosphataemia should be managed initially with intravenous hydration. Phosphate binders can be used to bind dietary phosphate; however, their contribution in phosphate reduction will be limited.

If hyperphosphataemia and hyperkalaemia are refractory to medical treatment, kidney replacement therapy (KRT) should be considered. Continuous cardiac monitoring may also be considered in some cases.

Treatment of hyperuricaemia

We recommend treatment of hyperuricaemia associated with TLS with rasburicase at a dose of 0.2 mg/kg/day.

This can be continued for up to 7 days (although 3–5 days is frequently sufficient) and the duration of treatment is determined by clinical response and uric acid levels. Allopurinol should be stopped when rasburicase is commenced and can be recommenced 24 h after the last dose of rasburicase.

We recognise that there is considerable interest in using lower doses, often reflecting cost considerations, with numerous studies reporting using reduced doses of rasburicase to treat TLS.

Many are retrospective studies that also include patients at high risk of TLS (i.e. prophylaxis rather than treatment) which complicates interpretation of the study results.^{44–46} One retrospective study⁴⁷ compared weight-based dosing to fixed doses of 7.5, 6 or 3 mg. There was no statistically significant difference between the groups in terms of the primary end-point of uric acid normalization at 24 h, but some patients required a further dose of rasburicase at 48 and 72 h. However, relatively few patients received fixed doses, for example, 3 mg (Laboratory TLS $n=18$; Clinical TLS $n=15$) compared to those who received weight-based doses (Laboratory TLS $n=126$; Clinical TLS $n=93$). Moreover, patients who received lower doses of rasburicase had lower uric acid levels, suggesting that the treating physicians only used low fixed doses of rasburicase in less severe cases of clinical TLS.

A few single-arm prospective clinical trials have reported the results of using fixed doses of rasburicase. Yaman et al.⁴⁸ used a single fixed 7.5 mg rasburicase dose in 36 patients with laboratory TLS and 46 patients with clinical TLS. Most cases occurred spontaneously rather than being treatment-related and all cases of clinical TLS were related to kidney injury. Seven patients in the cohort required haemodialysis and one patient remained dialysis dependent after treatment. Majumdar et al.⁴⁹ gave 1.5 mg rasburicase to 61 patients (50 laboratory TLS; 11 clinical TLS). Eleven patients had kidney impairment, of which three required haemodialysis. All patients recovered renal function. Other studies have suggested using fixed doses of rasburicase that are stratified by weight (e.g. 30–60 kg 3 mg; 60–90 kg 4.5 mg; >90 kg 6 mg)⁵⁰ or body mass index (BMI) (Laboratory TLS: 3 mg [BMI <30]; 4.5 mg [BMI >30] and clinical TLS: 6 mg [BMI <30]; 7.5 mg [BMI >30])⁵¹ albeit in smaller cohorts of patients. Interestingly, one reported only 57% adherence to protocol, which may relate to the complexity of their weight-stratified dosing protocol.⁵¹ This highlights the importance of a clear consistent dosing policy to improve adherence.

In the paediatric setting, there is a lack of prospective studies examining whether alterations in the recommended dose of rasburicase can be used to treat established TLS, without leading to decreased efficacy. A few retrospective studies from low-middle income countries have been reported, as the cost of rasburicase can be prohibitive in resource-limited situations. Appaji et al.⁵² used a fixed dose of 1.5 mg of rasburicase to treat 22 children with established TLS (6 clinical TLS; 16 laboratory TLS). Of 22 children, 20 responded with uric acid <8 mg/dL at 24 h, while the remaining two children

required a further dose of rasburicase and responded following the second dose.

However, other studies reported reduced efficacy, with only 75%⁵³ and 77.7%⁵⁴ of patients successfully normalising uric acid levels with lower doses of rasburicase. One study reported 2 of 18 children suffering sudden death from spontaneous TLS, and this approach cannot be recommended in a clinical setting where adequate resources are available.

Despite the interest in using doses of rasburicase that are lower than that recommended by the manufacturer, there is a lack of randomised clinical trial data to recommend this approach.

In very unwell, acutely presenting patients, consideration should be made to give rasburicase without waiting for a G6PD result. This decision should be made by the consultant with overall patient responsibility and following thorough evaluation of the risks and benefits for the individual patient.

Regular clinical and blood monitoring

It is important to perform repeated measurement of serum potassium, uric acid, phosphate, calcium and creatinine. The full set of blood tests is required every 6–12 h. However, in life-threatening electrolyte disturbances, more frequent targeted electrolyte monitoring (e.g. potassium in life-threatening hyperkalaemia) may support clinical decision-making. Regular clinical reassessment of the patient's fluid balance status and urine output is also required to monitor response to therapy and guide ongoing management.

Indications for KRT

Indications for KRT in TLS are in line with those for other causes of acute kidney injury (AKI) and include hyperkalaemia, fluid overload and severe metabolic acidosis that is refractory to medical management. However, the threshold for KRT should be lower compared to other clinical pathologies as tumour cell breakdown is ongoing and the rises in serum electrolytes may be rapid and unpredictable. As a result, in the context of AKI and/or significant reduction in urine output, hyperuricaemia, hyperphosphataemia and hypocalcaemia that are refractory to medical management should prompt early discussion with nephrology or intensive care physicians for timely consideration of KRT.

There is no available evidence to guide choice of intermittent versus continuous KRT. Continuous KRT in an intensive care setting should be considered in those patients who are critically unwell or display haemodynamic instability.

Recommendations

- **Patients with TLS should have vigorous hydration with intravenous fluid, at an initial rate of 3 L/m²/day for**

adults and up to 4 L/m²/day for children, depending on renal function, clinical and fluid status. The aim is to maintain a urine output of 100 mL/m²/h in adults (1B).

- Treatment of electrolyte disturbances in TLS should be initiated and managed as per local guidelines and protocols. If refractory to treatment, KRT should be considered. The threshold for KRT may be lower than other clinical pathologies as tumour cell breakdown is ongoing and the rises in serum electrolytes may be rapid and unpredictable (1B).
- Adult and paediatric patients with established TLS (laboratory or clinical TLS) should be treated with rasburicase 0.2 mg/kg/day for up to 7 days (although 3–5 days is frequently sufficient) with the exact duration based upon clinical response (1B).
- Allopurinol should be stopped when rasburicase is commenced and can be recommenced 24 h after the last dose of rasburicase (2C).
- In very unwell, acutely presenting patients, consideration may be made to give rasburicase without waiting for a G6PD result. This decision should be made by the consultant with overall patient responsibility and following thorough evaluation of the risks and benefits (2C).
- Serum potassium, uric acid, phosphate, calcium and creatinine should be measured every 6–12 h. More frequent targeted electrolyte monitoring may be required to support decision-making (2C).
- Regular clinical reassessment of the patient's fluid balance status and urine output is required to guide ongoing management (2C).

PRACTICAL CONSIDERATIONS FOR TLS MANAGEMENT—WITH EMPHASIS ON SYSTEM RISKS AND PATIENT SAFETY

There have been serious incidents of harm where patients with haematological malignancies have experienced delays in receiving medication prescribed to prevent TLS. We have collaborated with the NHSE Patient Safety team and SPS to identify the contributory factors and mitigate against them. The resulting SPS resource 'Ensuring time critical use of rasburicase' is available on the SPS website.⁵⁵ A summary of recommendations is listed below.

Recommendations: Drug availability/access

- Allopurinol must be available in all hospital settings where it may be required (1B).
- Rasburicase (requires refrigeration) should be made available in any clinical setting which regularly treats patients who could require either prophylactic or treatment doses of rasburicase (1B).
- Rasburicase should be available for patient administration within 1 h of prescription (1B).

- All centres should consider the availability of rasburicase in emergency supply fridges (depending on local arrangements of access to emergency medicines at the centre as some hospitals may have an on-site out of hours pharmacist). Both strengths of rasburicase (1.5 and 7.5 mg/mL) should be made available and easily accessible. Stock levels should be maintained at a sufficient level according to local policy and clinical workload (1B).
- Healthcare professionals in all clinical settings across the organisation should have an awareness and access to information to support timely access to rasburicase. There may be organisational policies/systems to support healthcare professionals in obtaining all medications required at the point of care (1B).

Recommendations: Prescribing and validation/screening

- Ensure that there is clarity in local policies and clinical guidelines/protocols regarding indications and appropriate dosing of rasburicase for the prophylaxis and treatment of TLS. This minimises the need for validation outside of the immediate clinical team (1B).
- Regular training is required for all staff involved in the care of patients at risk of TLS or who have developed TLS (1C).
- Ideally, patients at high risk of TLS should be under the care of a haematologist with experience of rasburicase use, TLS management and managed on specialist wards with experience in managing TLS, for example, haematology, intensive care, renal wards (2C).
- If patients at high risk of TLS are managed in non-specialist settings, it is important to ensure that patients receive the recommended monitoring and therapies for TLS and their haematological malignancy. Resources should be in place for at least twice daily clinical review of patients including review of laboratory results. The haematology team should provide medical, nursing and pharmacy oversight to support management (2C).
- Where rasburicase is used in a non-specialist setting, organisational policy should ensure that risks associated with delayed treatment are highlighted, and additional information and support provided on each occasion (2B).
- Ensure healthcare professionals responsible for the prescribing, supply and administration of rasburicase are aware of the risks associated with delays to treatment and when to escalate concerns (1B).
- Care must be taken to ensure that, where separate prescribing systems are in use, TLS prophylaxis/treatment is consistently prescribed on the most appropriate system. This is particularly important when patients are transferred from one clinical area to another (e.g. day unit setting to a ward) (1B).

CONCLUSION

TLS remains an important cause of morbidity and mortality in adults and children with a haematological malignancy. The 2015 BSH guidelines on the management of TLS remain relevant, and overall strategies in prophylaxis and treatment remain largely unchanged. Importantly, febuxostat is now available for TLS prophylaxis in low- or intermediate-risk patients. This may be useful for patients who are allergic or have skin reactions to allopurinol although it should be noted there have been MHRA warnings about its use in patients with major cardiovascular disease.

There is an increasing body of evidence to endorse the use of a fixed 3 mg dose of rasburicase for prophylaxis in high-risk adult patients. However, there is insufficient evidence to recommend dose reductions for the treatment of clinical TLS and so we continue to recommend rasburicase 0.2 mg/kg/day for up to 7 days for adults and children.

Rapid advances in therapeutics have been of undoubted benefit to our patients, but highly effective novel agents causing rapid cell breakdown can increase the risk of TLS, which needs to be recognised and managed effectively to ensure patient safety. Consequently, risk stratification is essential for all patients with haematological malignancy receiving anti-cancer therapy. This requires meticulous consideration of patient, disease and therapy factors to determine a personalised approach in TLS prophylaxis.

Despite this, cases of patient harm from delayed administration of TLS therapy occur and frequently reflect systemic risks in a healthcare system when specialist medications are used in unfamiliar settings. This updated guideline has been developed in conjunction with the NHSE Patient Safety team and the SPSs to identify these risks and provide additional recommendations to mitigate against them.

AUTHOR CONTRIBUTIONS

All authors contributed to writing and critically appraising this guideline.

ACKNOWLEDGEMENTS

The authors wish to thank Ben Stevenson for help in undertaking the initial literature review. The authors would also like to thank Pauline Lockey and Graeme Kirkpatrick (NHSE Patient Safety team) and Anna Bischler (Specialist Pharmacy Services) for their assistance in the writing of this guideline. The BSH Haemato-oncology task force members at the time of writing this guideline were Nilima Parry-Jones, Toby Eyre, Simon Stern, Andrew Clark, Andrew McGregor, Jo Ewing, Austin Kulasekaraj, Dima El-Sharkawi, Matthew Wilson, Rob Sellar and Jahanzaib Khwaja. The authors would like to thank them, the BSH sounding board and the BSH guidelines committee for their support in preparing this guideline. YLTC formed and led the writing group on behalf of the BSH Haemato-oncology taskforce.

CONFLICT OF INTEREST STATEMENT

The BSH paid the expenses incurred during the writing of this guidance. All authors have made a declaration of interests to the BSH and Task Force Chairs, which may be reviewed on request. YLTC has declared consultancy fees for Aptitude Health. DE-S has declared honoraria from Abbvie, AstraZeneca, Beigene, Gilead, Janssen, Lilly, Roche, Takeda; conference/travel support from Abbvie, Novartis, Roche; is on the advisory board for Abbvie, ASTEX, AstraZeneca, Beigene, Janssen, Kyowa Kirin, Roche and is a member of The Royal Marsden NHS Foundation Trust. SDM is a Professor of Paediatric Nephrology and Transplantation and Director of the NIHR GOSH Clinical Research Facility, so his institution receives funding for clinical trials from Novartis, Astellas, GSK and Aurinia. DC is a current study investigator and previous investigator with MSD. GJ has received an educational grant from Abbvie to facilitate conference attendance. The following members of the writing group: DO'C and PS have no conflicts of interest to declare.

REVIEW PROCESS

Members of the writing group will inform the writing group Chair if any new pertinent evidence becomes available that would alter the strength of the recommendations made in this document or render it obsolete. The document will be archived and removed from the BSH current guidelines website if it becomes obsolete. If new recommendations are made, an addendum will be published on the BSH guidelines website (Guidelines [[b-s-h.org.uk](https://www.bsh.org.uk)]).

DISCLAIMER

While the advice and information in this guidance are believed to be true and accurate at the time of going to press, neither the authors, the BSH nor the publishers accept any legal responsibility for the content of this guidance.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Chan YLT, El-Sharkawi D, Shah P, Chanouzas D, O'Connor D, Marks SD, et al. British Society for Haematology updated guidelines for the diagnosis and management of tumour lysis syndrome in adults and children with haematological malignancies: A focus on patient safety. *Br J Haematol*. 2025;207(4):1248–1258. <https://doi.org/10.1111/bjh.70092>